

JAVA Vityarthi BYOP BY 25BAI11043

Student Performance Management System

 Java 17+  Dependencies None  License Academic  Status Active

Overview

I built this project to make student record and marks management easier without having to calculate results manually every time. It is a Java application operated from a console which keeps track of student information, stores the marks according to subject, and automatically computes the percentage, the grade, and the pass/fail status.

- Modules: **Student Management, Marks Management, Performance Reports**
- Storage: **CSV file** (`students.csv`) for storing student records
- Validation: **Custom checked exceptions** for handling invalid input
- Testing: **16 test checks**, all passing

Key Features

- Add, update, delete, and search students using ID or name
- Enter subject-wise marks with **0–100 validation**
- Automatically calculates **percentage and letter grade**
- Shows **class average, highest/lowest scorer, and pass/fail results**
- The changes are automatically saved and the student data is automatically reloaded from `students.csv`
- Separate tests are written for **Student, StudentManager, and FileManager**

Technologies / Tools Used

- **Language:** Java (JDK 17+, built and tested on JDK 21)
- **Build:** plain `javac` / `java` no Maven, no Gradle
- **Storage:** flat-file CSV (`students.csv`) no database
- **Testing:** a hand-rolled test harness (`AppTest.java`) no JUnit or other framework
- **Version control:** Git / GitHub

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Project Structure

```
Vityarthi_Java_Project_25BAI11043/
├── Diagrams/
│   ├── architecture-diagram.png
│   ├── class-diagram.png
│   ├── sequence-diagram.png
│   ├── usecase-diagram.png
│   └── workflow-diagram.png
├── Screenshots/
│   ├── add-student.png
│   ├── class-average.png
│   ├── delete-student.png
│   ├── enter-marks.png
│   ├── highest-lowest.png
│   ├── individual-report.png
│   ├── main-menu.png
│   ├── pass-fail.png
│   ├── percentage-grade.png
│   ├── save-load.png
│   ├── search-student.png
│   ├── test-run.png
│   └── update-student.png
├── out/
│   ├── AppTest.class
│   ├── FileManager.class
│   ├── InvalidMarksException.class
│   ├── Main.class
│   ├── PerformanceReport.class
│   ├── Student.class
│   ├── StudentManager.class
│   └── StudentNotFoundException.class
├── src/
│   └── Student.java                                # One student's data + grade/percentage
math
│   ├── StudentManager.java                        # CRUD on the in-memory roster
│   └── PerformanceReport.java                     # Individual + class-level report
generation
│   └── FileManager.java                           # CSV save/load
```

```

|   |─ InvalidMarksException.java    # Custom checked exception - bad marks
input
|   |─ StudentNotFoundException.java # Custom checked exception - bad ID lookup
|   |─ Main.java                    # Console menu / entry point
|   |─ AppTest.java                 # Dependency-free test harness
|─ .gitignore
|─ Project_report.pdf               # Detailed report, submitted separately on
the portal
|─ README.md
|─ README.pdf
|─ statement.md                    # Problem statement, scope, target users
|─ students.csv

```

`out/` is just what `javac` spits out locally, I'm not committing it. `.gitignore` keeps `out/`, `bin/` and `*.class`, out of the repo, so only `src/`, `diagrams/`, the docs and `students.csv` (included with sample data so the app runs with real records out of the box) actually get pushed.

Requirements

Requirement	Version
JDK	17+ (I built and tested this on JDK 21)
Build tool	None just <code>javac</code> / <code>java</code>
External libraries	None

- **Hardware:** anything that can run a JVM
- **Storage:** basically nothing `students.csv` is a few bytes per student

Non-Functional Requirements

1. **Performance** : CRUD operations and report generation complete in under 1 second for a roster of a few hundred students, since everything is held in memory as an `ArrayList<Student>`.
2. **Reliability** : Every add, update, delete, or marks entry is immediately persisted to `students.csv`, so no data is lost if the program exits unexpectedly.
3. **Usability** : A numbered console menu (0–10) guides the user through every operation with clear prompts, so no prior documentation is needed to operate the system.
4. **Maintainability** : Each responsibility is isolated into its own class (`Student`, `StudentManager`, `FileManager`, `PerformanceReport`), so a change to one concern (e.g. storage format) doesn't require touching the others.
5. **Error Handling / Robustness** : Custom checked exceptions (`InvalidMarksException`, `StudentNotFoundException`) force calling code to handle bad input explicitly rather than failing silently; a corrupted CSV line is skipped with a warning instead of crashing the whole load.
6. **Resource Efficiency** : No external dependencies, no database server, and no build tool overhead, the entire system runs on a bare JDK install with a CSV file a few bytes in size.

Installation

1. Clone it:

```
git clone
https://github.com/iampiyushsingh/Vityarthi_Java_Project_25BAI11043.git
cd Vityarthi_Java_Project_25BAI11043
```

2. Check your Java version:

```
java -version
javac -version
```

Both `java` and `javac` should report version 17 or higher.

```
javac -d out src/*.java
```

This compiles all Java source files in `src/` and places the generated `.class` files in `out/`. If no compilation errors are displayed, the project has compiled successfully.

Configuration

The project does not require any special setup or configuration. Just run the program and start adding student records. The required `students.csv` file is created automatically when the first student is added.

Usage

Run it

```
java -cp out Main
```

which drops you into:

```
===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice:
```

Type the number which corresponds to the operation you want to carry out and then carry out the instructions shown on the screen. Choose 0 if you want to save the data and exit the programme. The application also automatically saves the changes after each add, update, delete or marking of an entry, thus helping to avoid data loss.

Run the tests

```
java -cp out AppTest
```

You should see this at the bottom:

```
===== TEST SUMMARY =====  
Passed: 16    Failed: 0
```

How It Works

Student Management (*StudentManager*)

Keeps a plain `ArrayList<Student>` in memory with IDs that auto-increment from 1. `addStudent`, `updateStudent`, `deleteStudent`, `findById`, `searchByName` cover the basics. Try to update or delete an ID that doesn't exist and you get a `StudentNotFoundException` instead of a silent no-op.

Marks Management (*Student*)

`addMarks(subject, marks)` checks the mark is between 0 and 100 and the subject isn't blank, anything else throws `InvalidMarksException`. Percentage is just total marks obtained divided by (number of subjects × 100). Grade comes off a fixed scale (A+ at 90%+, down to F below 40%), and 40% is the pass line.

Performance Reports (*PerformanceReport*)

- `individualReport` : one student's full report card
- `classAverage` : mean percentage across everyone
- `highestLowest` : top and bottom scorer
- `passFailAnalysis` : who passed, who didn't, and the class pass rate

Persistence (*FileManager*)

Each student becomes one CSV line: `id,name,subject:marks,subject:marks,...`. On startup it reads the file back, rebuilds every `Student`, and even restores `nextId` correctly so new students don't collide with old IDs. If a line's gotten corrupted somehow, it skips that record and prints a warning instead of crashing the whole load.

Data File

File	Purpose
<code>students.csv</code>	The whole roster + marks, auto-generated. Delete it if you want to start over.

Workflow

```
Console input (Main) → StudentManager (CRUD) / Student (marks, grade, %)  
                        ↓  
                FileManager saves after every change  
                        ↓  
                students.csv persists between runs  
                        ↓  
                PerformanceReport (average, highest/lowest, pass/fail)
```

Testing

`AppTest.java` doesn't use any testing framework, just a `check()` helper and plain `main()`. It covers:

- Add / find / update / delete, including the exception path when the ID doesn't exist
- Grade and percentage math
- Rejecting out-of-range and negative marks
- Class average and pass/fail logic
- A full save-then-reload round trip through `FileManager`

Run it with `java -cp out AppTest`.

Screenshots

Main Menu

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>  
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2  
  
===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====  
1. Add Student  
2. Update Student  
3. Delete Student  
4. Search Student  
5. Enter Marks  
6. View Percentage & Grade  
7. Individual Report  
8. Class Average  
9. Highest/Lowest Marks  
10. Pass/Fail Analysis  
0. Save & Exit  
Enter your choice:
```

Add Student

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 1
Enter student name: Piyush
Student added successfully! Assigned ID = 1
```

Update Student

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
Messages' -cp' 'C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\Main'
Loaded 1 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 2
Enter student ID to update: 1
Enter new name: Ayush
Student updated successfully!
```

Delete Student

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
Messages' '-cp' 'C:\Users\piyus\AppData\Roaming\Code\User\workspaceSto
Main'
Loaded 1 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 3
Enter student ID to delete: 1
Student deleted successfully!
```

Search Student

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
Messages' '-cp' 'C:\Users\piyus\AppData\Roaming\Code\User\workspaceSto
Main'
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 4
Search by: 1) ID 2) Name
Choice: 1
Enter ID: 1
ID: 1    Name: Piyush          Percentage: 97.67% Grade: A+
```

Enter Marks


```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 5
Enter student ID: 3
Enter subject name: Maths
Enter marks obtained (0-100): 98
Marks recorded for Arghav in Maths
```

Percentage and Grade

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 6
Enter student ID: 3
Percentage: 94.67% Grade: A+
```

Individual Report

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
Messages' '-cp' 'C:\Users\piyus\AppData\Roaming\Code\User\workspaceSto
Main'
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 7
Enter student ID: 2

----- Individual Report -----
ID   : 2
Name : Ayush
Subject-wise marks:
  Maths      : 90/100
  English    : 86/100
  Hindi      : 92/100
Total       : 268 / 300
Percentage  : 89.33%
Grade      : A
Result     : PASS
-----
```

Class Average

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 8

Class Average Percentage: 93.89% (based on 3 students)
```

Highest and Lowest Marks

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -Xmx1024m -Xms1024m -Duser.timezone=Asia/Kolkata -Duser.language=en -Duser.country=IN -Duser.variant=IN -Duser.home=C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf25cf68798862f0
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 9

Highest Scorer: ID: 1      Name: Piyush      Percentage: 97.67% Grade: A+
Lowest Scorer : ID: 2      Name: Ayush       Percentage: 89.33% Grade: A
```

Pass Fail Analysis

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -Xmx1024m -Xms1024m -Duser.timezone=Asia/Kolkata -Duser.language=en -Duser.country=IN -Duser.variant=IN -Duser.home=C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 10

----- Pass/Fail Analysis -----
Total Students : 3
Passed         : 3
Failed         : 0
Pass Rate      : 100.00%
```

Save and Load

```
PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
● C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
Loaded 3 student(s) from students.csv

===== STUDENT PERFORMANCE MANAGEMENT SYSTEM =====
1. Add Student
2. Update Student
3. Delete Student
4. Search Student
5. Enter Marks
6. View Percentage & Grade
7. Individual Report
8. Class Average
9. Highest/Lowest Marks
10. Pass/Fail Analysis
0. Save & Exit
Enter your choice: 0
Data saved. Exiting... Goodbye!
```

Test Run

```
● PS C:\Users\piyus\OneDrive\Desktop\Vityarthi_Java_Project_25BAI11043>
C:\Users\piyus\AppData\Roaming\Code\User\workspaceStorage\bec93fbacaf2
[PASS] addStudent assigns ID 1 to first student
[PASS] findById locates the newly added student
[PASS] updateStudent changes the name
[PASS] deleteStudent removes the student
[PASS] updateStudent throws StudentNotFoundException for a bad ID
[PASS] Percentage calculated correctly (95+85)/200
[PASS] Grade for 90% is A+
[PASS] isPass() is true for 90%
[PASS] Marks > 100 are rejected
[PASS] Negative marks are rejected
[PASS] Class average is (80+20)/2 = 50%
[PASS] Student A passes
[PASS] Student B fails
[PASS] Reloaded manager has 1 student
[PASS] Reloaded student has correct name
[PASS] Reloaded student has correct marks

===== TEST SUMMARY =====
Passed: 16   Failed: 0
```

Troubleshooting

Problem	Fix
'javac' is not recognized / command not found: javac	JDK isn't installed, or isn't on your PATH, install JDK 17+.
error: release version 21 not supported	The installed JDK is older than JDK 21. Check <code>java -version</code> and <code>javac -version</code> .
Could not find or load main class Main	Run the command from the project root, and make sure <code>out/Main.class</code> actually exists (compile first).

Problem	Fix
Program exits or skips a prompt right away	Type a plain number with no stray spaces, then press Enter.
Data isn't there next time I run it	Run <code>java -cp out Main</code> from the exact same folder each time <code>students.csv</code> is relative to wherever you launch it from.
Warning: skipped corrupted record	Someone (probably me) hand-edited <code>students.csv</code> and broke a line fix or delete that line.

Notes

Project Notes

- The data for all students is kept in one CSV file, which means that no setup is needed, no server is required and no additional software is necessary apart from the JDK.
- Reports are produced using the student data which has been loaded from the file `students.csv` when the program is started.
- The files `out/`, `bin/`, `*.class`, and `students.csv` are included in `.gitignore` since they are generated files and therefore not part of the source code.
- The full project report, with the diagrams and information about the testing, is included in `Project_Report.pdf`.

Key Classes & Methods

- `Student.addMarks()` : validates and records one subject's marks
- `Student.getPercentage()` / `getGrade()` / `isPass()` : the derived numbers
- `StudentManager.addStudent()` / `updateStudent()` / `deleteStudent()` : roster CRUD
- `StudentManager.getIdOrThrow()` : lookup that throws instead of returning null
- `FileManager.save()` / `load()` : the CSV round trip
- `PerformanceReport.individualReport()` / `classAverage()` / `highestLowest()` / `passFailAnalysis()` : the reporting side

License

This project is part of an AIML course assignment at VIT Bhopal University.

Contact

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